WEEK 1

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Course Title : Problem Solving And Testing Using Java

Course code :10211CS227

Session 1: Introduction to Functional Programming Concepts

Java Lambda Expressions:

Write the following methods that *return a lambda expression* performing a specified action:

1. PerformOperation isOdd(): The lambda expression must return if a number is odd or if it is even.
2. PerformOperation isPrime(): The lambda expression must return if a number is prime or if it is composite.
3. PerformOperation isPalindrome(): The lambda expression must return if a number is a palindrome or if it is not.

Input Format

Input is handled for you by the locked stub code in your editor.

Output Format

The locked stub code in your editor will print lines of output.

Sample Input

The first line contains an integer, (the number of test cases).

The subsequent lines each describe a test case in the form of space-separated integers:
The first integer specifies the condition to check for (for Odd/Even, for Prime, or for Palindrome). The second integer denotes the number to be checked.

5

1 4

2 5

3 898

1 3

2 12

Sample Output

EVEN

PRIME

PALINDROME

ODD

COMPOSITE

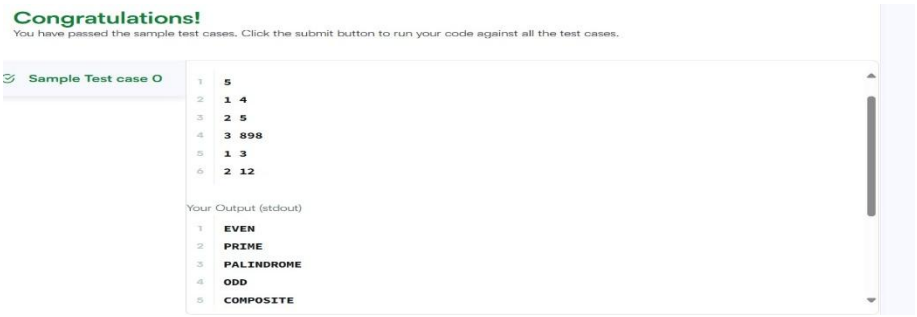
Program:

```
import java.io.*;
import java.util.*;
interface PerformOperation {
 boolean check(int a);
}
class MyMath {
 public static boolean checker(PerformOperation p, int num) {
  return p.check(num);
 }

   public PerformOperation isOdd() {
      return n -> n % 2 != 0;
   }

   public PerformOperation isPrime() {
      return n -> {
         if (n < 2)
            return false;

         for (int i = 2; i * i <= n; i++) {
            if (n % i == 0)
               return false;
         }
```

```
        return true;
      };
    }

    public PerformOperation isPalindrome() {
      return n -> {
        int original = n;
        int reverse = 0;

        while (n > 0) {
          reverse = reverse * 10 + n % 10;
          n /= 10;
        }

        return original == reverse;
      };
    }
}
public class Solution {

 public static void main(String[] args) throws IOException {
  MyMath ob = new MyMath();
  BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
  int T = Integer.parseInt(br.readLine());
  PerformOperation op;
  boolean ret = false;
  String ans = null;
  while (T--> 0) {
```

```
  String s = br.readLine().trim();
  StringTokenizer st = new StringTokenizer(s);
  int ch = Integer.parseInt(st.nextToken());
  int num = Integer.parseInt(st.nextToken());
  if (ch == 1) {
   op = ob.isOdd();
   ret = ob.checker(op, num);
   ans = (ret) ? "ODD" : "EVEN";
  } else if (ch == 2) {
   op = ob.isPrime();
   ret = ob.checker(op, num);
   ans = (ret) ? "PRIME" : "COMPOSITE";
  } else if (ch == 3) {
   op = ob.isPalindrome();
   ret = ob.checker(op, num);
   ans = (ret) ? "PALINDROME" : "NOT PALINDROME";

  }
  System.out.println(ans);
 }
}
}
```

Output:



Session 2: Lambda Expressions

Java Sort:

You are given a list of student information: ID, FirstName, and CGPA. Your task is to rearrange them according to their CGPA in decreasing order. If two student have the same CGPA, then arrange them according to their first name in alphabetical order. If those two students also have the same first name, then order them according to their ID. No two students have the same ID.

Hint: You can use comparators to sort a list of objects. See the oracle docs to learn about comparators.

Input Format

The first line of input contains an integer , representing the total number of students. The next lines contains a list of student information in the following structure:

ID Name CGPA

Output Format

After rearranging the students according to the above rules, print the first name of each student on a separate line.

Sample Input

5

33 Rumpa 3.68

85 Ashis 3.85

56 Samiha 3.75

19 Samara 3.75

22 Fahim 3.76

Sample Output

Ashis

Fahim

Samara

Samiha

Rumpa

Program:

```
import java.util.*;


class Student{
	private int id;
	private String fname;
	private double cgpa;
	public Student(int id, String fname, double cgpa) {
		super();
		this.id = id;
		this.fname = fname;
		this.cgpa = cgpa;
	}
	public int getId() {
		return id;
	}
	public String getFname() {
		return fname;
	}
	public double getCgpa() {
		return cgpa;
	}
}

//Complete the code
public class Solution
{
	public static void main(String[] args){
```

```
                Scanner in = new Scanner(System.in);
                int testCases = Integer.parseInt(in.nextLine());
                List<Student> studentList = new ArrayList<Student>();
                while(testCases>0){
                        int id = in.nextInt();
                        String fname = in.next();
                        double cgpa = in.nextDouble();
                        Student st = new Student(id, fname, cgpa);
                        studentList.add(st);
                        testCases--;
                }
    Collections.sort(studentList, Comparator
      .comparing(Student::getCgpa, Comparator.reverseOrder())
      .thenComparing(Student::getFname)
      .thenComparing(Student::getId));
       for(Student st: studentList){
                        System.out.println(st.getFname());
                }
        }
}
```

Output:

Sort Array By Parity:

Given an integer array nums, move all the even integers at the beginning of the array followed by all the odd integers.

Return ***any array*** *that satisfies this condition*.

Example 1:

Input: nums = [3,1,2,4]

Output: [2,4,3,1]

Explanation: The outputs [4,2,3,1], [2,4,1,3], and [4,2,1,3] would also be accepted.

Example 2:

Input: nums = [0]

Output: [0]

Constraints:

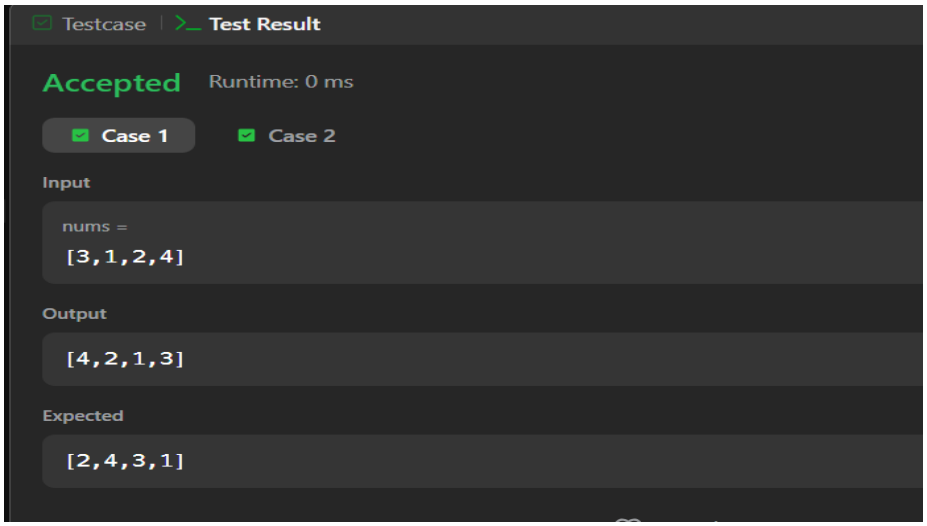
- 1 <= nums.length <= 5000
- 0 <= nums[i] <= 5000

Program:

```
class Solution {
  public int[] sortArrayByParity(int[] nums) {
    int left = 0, right = nums.length - 1;

    while (left < right) {
      if (nums[left] % 2 > nums[right] % 2) {
        int temp = nums[left];
        nums[left] = nums[right];
        nums[right] = temp;
```

```
        }

        if (nums[left] % 2 == 0) left++;
        if (nums[right] % 2 == 1) right--;
      }

      return nums;
    }
}
```

Output:



Session 3: Anonymous Functions & Functional Interfaces

Sorting: Comparator:

Comparators are used to compare two objects. In this challenge, you'll create a comparator and use it to sort an array. The *Player* class is provided in the editor below. It has two fields:

1. : a string.
2. : an integer.

Given an array of *Player* objects, write a comparator that sorts them in order of decreasing score. If or more players have the same score, sort those players alphabetically ascending by name. To do this, you must create a *Checker* class that

implements the *Comparator* interface, then write an *int compare(Player a, Player b)* method implementing the Comparator.compare(T o1, T o2) method. In short, when sorting in ascending order, a comparator function returns if , if , and if .

Declare a *Checker* class that implements the *comparator* method as described. It should sort first descending by score, then ascending by name. The code stub reads the input, creates a list of Player objects, uses your method to sort the data, and prints it out properly.

Example

Sort the list as . Sort first descending by score, then ascending by name.

Input Format

The first line contains an integer, , the number of players.
Each of the next lines contains a player's and , a string and an integer.

Constraints

- Two or more players can have the same name.
- Player names consist of lowercase English alphabetic letters.

Output Format

You are not responsible for printing any output to stdout. Locked stub code in *Solution* will instantiate a *Checker* object, use it to sort the *Player* array, and print each sorted element.

Sample Input

5

amy 100

david 100

heraldo 50

aakansha 75

aleksa 150

Sample Output

aleksa 150

amy 100

david 100

aakansha 75

heraldo 50

Program:

```
import java.util.*;


class Player {
        String name;
        int score;

        Player(String name, int score) {
                this.name = name;
                this.score = score;
        }
}

class Checker implements Comparator<Player> {
        // complete this method
        public int compare(Player a, Player b) {
    if (a.score != b.score) {
      // Sort score in descending order
      return Integer.compare(b.score, a.score);
    }
    // If scores are equal, sort name in ascending order (alphabetical)
    return a.name.compareTo(b.name);
  }
}
public class Solution {
```

```
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int n = scan.nextInt();
        Player[] player = new Player[n];
        Checker checker = new Checker();
        for(int i = 0; i < n; i++){
            player[i] = new Player(scan.next(), scan.nextInt());
        }
        scan.close();
        Arrays.sort(player, checker);
        for(int i = 0; i < player.length; i++){
            System.out.printf("%s %s\n", player[i].name, player[i].score);
        }
    }
}
```

Output:

Java Comparator:

Comparators are used to compare two objects. In this challenge, you'll create a comparator and use it to sort an array.

The *Player* class is provided for you in your editor. It has fields: a String and a integer.

Given an array of *Player* objects, write a comparator that sorts them in order of decreasing score; if or more players have the same score, sort those players alphabetically by name. To do this, you must create a *Checker* class that implements the *Comparator* interface, then write an *int compare(Player a, Player b)* method implementing the Comparator.compare(T o1, T o2) method.

Input Format

Input from stdin is handled by the locked stub code in the *Solution* class.

The first line contains an integer, , denoting the number of players.
Each of the subsequent lines contains a player's and , respectively.

Constraints

- players can have the same name.
- Player names consist of lowercase English letters.

Output Format

You are not responsible for printing any output to stdout. The locked stub code in *Solution* will create a *Checker* object, use it to sort the *Player* array, and print each sorted element.

Sample Input

5

amy 100

david 100

heraldo 50

aakansha 75

aleksa 150

Sample Output

aleksa 150

amy 100

david 100

aakansha 75

heraldo 50

Program:

```
import java.util.*;
class Checker implements Comparator<Player> {
  @Override
  public int compare(Player a, Player b) {
    if (a.score != b.score) {
      return b.score - a.score; // Decreasing order of score
    }
    return a.name.compareTo(b.name); // Alphabetical order of name
  }
}

class Player{
  String name;
  int score;

  Player(String name, int score){
    this.name = name;
    this.score = score;
  }
}

class Solution {

  public static void main(String[] args) {
    Scanner scan = new Scanner(System.in);
    int n = scan.nextInt();
```

```
        Player[] player = new Player[n];
        Checker checker = new Checker();

        for(int i = 0; i < n; i++){
            player[i] = new Player(scan.next(), scan.nextInt());
        }
        scan.close();

        Arrays.sort(player, checker);
        for(int i = 0; i < player.length; i++){
            System.out.printf("%s %s\n", player[i].name, player[i].score);
        }
    }
}
```

Output:

Session 4: Higher-Order Functions

Running Sum of 1D Array

Given an array nums. We define a running sum of an array as runningSum[i] = sum(nums[0]...nums[i]).

Return the running sum of nums.

Example 1:

Input: nums = [1,2,3,4]

Output: [1,3,6,10]

Explanation: Running sum is obtained as follows: [1, 1+2, 1+2+3, 1+2+3+4].

Example 2:

Input: nums = [1,1,1,1,1]

Output: [1,2,3,4,5]

Explanation: Running sum is obtained as follows: [1, 1+1, 1+1+1, 1+1+1+1, 1+1+1+1+1].

Example 3:

Input: nums = [3,1,2,10,1]

Output: [3,4,6,16,17]

Constraints:

- 1 <= nums.length <= 1000
- -10^6 <= nums[i] <= 10^6

Program:

```
class Solution {
   public int[] runningSum(int[] nums) {
      for (int i = 1; i < nums.length; i++) {
         nums[i] += nums[i - 1];
```

```
    }
    return nums;
  }
}
```

Output:

Richest Customer Wealth:

You are given an m x n integer grid accounts where accounts[i][j] is the amount of money the i^{th} customer has in the j^{th} bank. Return *the wealth that the richest customer has*.

A customer's wealth is the amount of money they have in all their bank accounts. The richest customer is the customer that has the maximum wealth.

Example 1:
Input: accounts = [[1,2,3],[3,2,1]]

Output: 6

Explanation:

1st customer has wealth = 1 + 2 + 3 = 6

2nd customer has wealth = 3 + 2 + 1 = 6

Both customers are considered the richest with a wealth of 6 each, so return 6.

Example 2:

Input: accounts = [[1,5],[7,3],[3,5]]

Output: 10

Explanation:

1st customer has wealth = 6

2nd customer has wealth = 10

3rd customer has wealth = 8

The 2nd customer is the richest with a wealth of 10.

Example 3:

Input: accounts = [[2,8,7],[7,1,3],[1,9,5]]

Output: 17

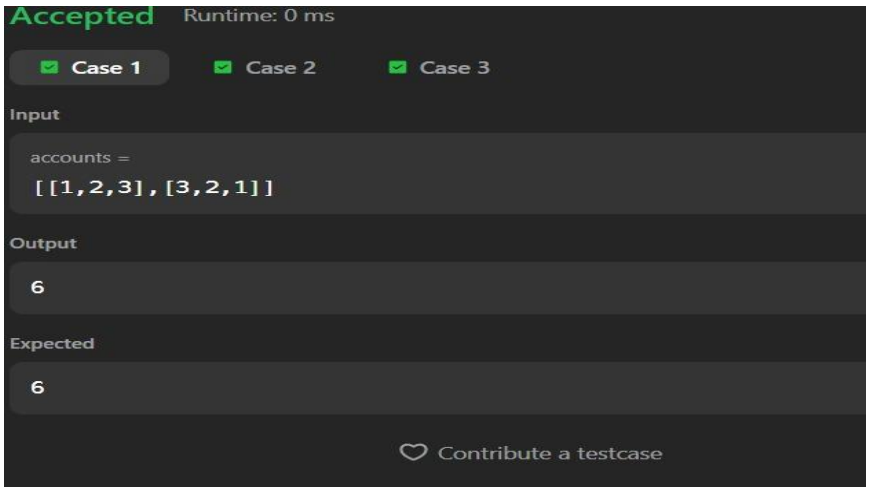
Constraints:

- m == accounts.length
- n == accounts[i].length
- 1 <= m, n <= 50
- 1 <= accounts[i][j] <= 100

Program:

```
class Solution {
    public int maximumWealth(int[][] accounts) {
        int maxWealth = 0;

        for (int[] customer : accounts) {
            int currentWealth = 0;
            for (int bank : customer) {
                currentWealth += bank;
            }
            maxWealth = Math.max(maxWealth, currentWealth);
        }

        return maxWealth;
    }
}
```

Output:



Session 5: Higher-Order Functions

Squares of a Sorted Array:

Given an integer array nums sorted in n

on-decreasing order, return *an array of the squares of each number sorted in non-decreasing order*.

Example 1:

Input: nums = [-4,-1,0,3,10]

Output: [0,1,9,16,100]

Explanation: After squaring, the array becomes [16,1,0,9,100].

After sorting, it becomes [0,1,9,16,100].

Example 2:

Input: nums = [-7,-3,2,3,11]

Output: [4,9,9,49,121]

Constraints:

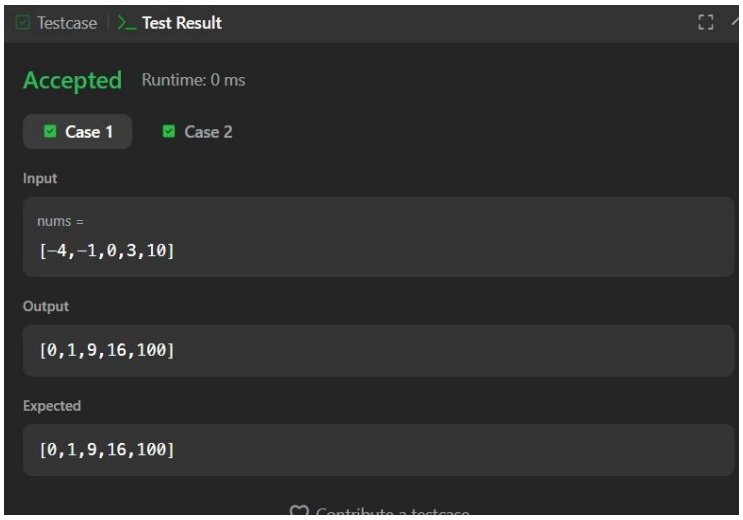
- 1 <= nums.length <= 10^4

- $-10^4 <= nums[i] <= 10^4$
- nums is sorted in non-decreasing order.

Program:

```
class Solution {
    public int[] sortedSquares(int[] nums) {
        int n = nums.length;
        int[] result = new int[n];

        int left = 0;
        int right = n - 1;
        int index = n - 1;

        while (left <= right) {
            if (Math.abs(nums[left]) > Math.abs(nums[right])) {
                result[index] = nums[left] * nums[left];
                left++;
            } else {
                result[index] = nums[right] * nums[right];
                right--;
            }
            index--;
        }

        return result;
    }
}
```

Output:



Find Pivot Index:

Given an array of integers nums, calculate the pivot index of this array.

The pivot index is the index where the sum of all the numbers strictly to the left of the index is equal to the sum of all the numbers strictly to the index's right.

If the index is on the left edge of the array, then the left sum is 0 because there are no elements to the left. This also applies to the right edge of the array.

Return *the leftmost pivot index*. If no such index exists, return -1.

Example 1:

Input: nums = [1,7,3,6,5,6]

Output: 3

Explanation:

The pivot index is 3.

Left sum = nums[0] + nums[1] + nums[2] = 1 + 7 + 3 = 11

Right sum = nums[4] + nums[5] = 5 + 6 = 11

Example 2:

Input: nums = [1,2,3]

Output: -1

Explanation:

There is no index that satisfies the conditions in the problem statement.

Example 3:

Input: nums = [2,1,-1]

Output: 0

Explanation:

The pivot index is 0.

Left sum = 0 (no elements to the left of index 0)

Right sum = nums[1] + nums[2] = 1 + -1 = 0

Constraints:

- 1 <= nums.length <= 10^4
- -1000 <= nums[i] <= 1000

Program:

```
class Solution {
  public int pivotIndex(int[] nums) {
    int totalSum = 0;
    for (int num : nums) {
      totalSum += num;
    }

    int leftSum = 0;
    for (int i = 0; i < nums.length; i++) {
      // Right sum is total sum minus left sum minus current element
      if (leftSum == totalSum - leftSum - nums[i]) {
        return i;
```

```
        }
        leftSum += nums[i];
      }

      return -1;
    }
}
```

Output: